Yuxuan Gu

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EDUCATION

Imperial College London MEng Electrical and Information Engineering **2021-2025**

- Dean's list for Year 2 (2022-2023)
- Dean's list for Year 1 (2021-2022) (top 5%)
- Modules: Circuits analysis, Analogue electronics, Digital electronics, Computer architecture, Control Systems, Signals and Communication, Programming, Mathematics
- Successfully completed all lab tasks, including challenging ones, and actively contributed to formulating project schedules and organizing implementation discussions with classmates.
- Demonstrated good time management skills, investigating literature involving control, robotics, electronics, and telecommunication in extra-curricular time

Jinling High School, Nanjing, China **2018-2021**

A Levels

- Accumulative marks Top 1 in China – November 2020 Cambridge Outstanding Learner Award (Best Across Three) and gave a speech as a Student Rep in the awards ceremony
- Further Mathematics (A*), Physics(A*), Chemistry(A*), Mathematics(A*)

WORK EXPERIENCE

Imperial College – Undergraduate Research Opportunities Programme (UROP) **07-09/2023**

- Developed a colour-tracking 4-DOF robotic arm using remote control within ROS2 framework
- Integrated a USB camera as a sensor within the feedback loop for effective bus servo motor management, while also employing geometric inverse kinematics analysis to calculate joint variables from end-effector position and orientation
- Employed a Raspberry Pi to transmit commands to the motors through UART, while concurrently publishing image frames to ROS2 topics
- Implemented controller separation onto a remote machine, enabling communication with the local node through the ROS2 network, and subsequently conducted an analysis of stability margins
- Built a 2nd order transfer function model by exploring step response of one motor

Imperial College – Undergraduate Teaching Assistant (UTA) **09/2023**

- Provided comprehensive assistance to first-year student during the lectures and problem classes

Evotrack – Software engineer summer internship **07-09/2022**

- Analysed usage data of E-vehicles charging stations in Paris and ran several k-means clustering algorithms to divide stations into clusters
- Established a routine to obtain the number of given list of facilities around given addresses using Google Maps API
- Trained ensemble models (e.g., Gradient Boosting) to predict the future utilisation of a station based on the past usage data and number of surrounding facilities and achieved high accuracy
- Developed excellent programming skills and machine learning knowledge by exploring all

Imperial College Careers Service Work Shadowing Scheme **08/2022**

- Experienced a day of life of an electrical engineer to maintain the water pumping systems
- Analysed pressure signal, PID feedback control of pump motor, and CHP units
- Compared parameters on the control panel in old-fashioned and modern plant rooms and discussed the development in this industry

PROJECTS

Imperial College London – Self-balancing autonomous maze-solving rover **06/2023**

- Developed a self-balancing rover that autonomously solves mazes, maps them in real-time, and identifies the shortest path
- Implemented a comprehensive system equipped with various components, such as NEMA 17 stepper motors, A4988 motor driver modules, an MPU-6050 gyroscope, an ESP32 microcontroller, an FPGA board, and a camera
- Leveraged FPGA vision technology to enhance image processing and analysis capabilities, enabling precise navigation and obstacle detection.
- Implemented sophisticated algorithms such as PID control, triangulation and A* algorithm, clustering methods, and graph theory to optimize rover performance and achieve efficient maze-solving strategies.
- Worked in a team of 2 EEE and 4 EIE people, assuming responsibility for FPGA vision, motor driving, PID control, and establishing seamless communication between esp32/PC and FPGA/esp32

Imperial College London – Remotely-controlled rover **06/2022**

- Designed a rover that can identify rocks emitting unusual electromagnetic and acoustic signals
- Utilised Fourier Transform, filtering methods and modulation & demodulation techniques
- Sharpened practical and programming skills by debugging faults in circuits and Arduino codes
- Collaborated with 5 teammates and responsible for all signal identifications

Light-seeking robot **02/2021**

- Successfully created a robot prototype that moves towards the light
- Carried out soldering PCB, programming to Arduino (C++), and testing the robot in a team of 2
- Employed electronic technology and motion control technique
- Demonstrated the functions to academics and tutor group members

AI-based plants water shortage warning device **07-09/2020**

- Developed research methodologies by exploiting online resources
- Developed a model to match the image of leaves with soil humidity
- Came up with the idea of using AI to remind ordinary people to water the plant
- Collected image of leaves and soil humidity data using Raspberry Pi and Arduino

Haihua AI Challenge -- Garbage Classification **02/2020**

- Achieved 4th place in the leader board
- Trained a CNN model to classify 80,000 rubbish images into 198 corresponding categories
- Distributed sub-tasks to 2 of my teammates and launched into model building and training using Python with the help of senior students and online resources

TECHNICAL SKILLS

- Programming language: C++, Python
- Simulation tools: MATLAB, LTSpice
- Microcontroller and embedded system: Arduino, Raspberry Pi
- Language: English (IELTS: overall 7.5 with each band no less than 7.0), Chinese (native)